

Local RAG Chatbot Project – Interview Questions & Simple Answers

1. What is your project?

Answer:

My project is a Local RAG-based Chatbot that answers questions from a PDF document using FAISS and LangChain.

2. What is RAG?

Answer:

RAG stands for **Retrieval-Augmented Generation**. It retrieves relevant information from documents and uses it to answer user questions.

3. Why did you use RAG?

Answer:

RAG helps provide accurate answers from my PDF document instead of relying only on the model's knowledge.

4. What is FAISS?

Answer:

FAISS is a vector database developed by Meta. It stores embeddings and performs fast similarity search.

5. What is an Embedding?

Answer:

An embedding is a numerical vector representation of text that captures its meaning.

Example:

"What is soybean?"

"Explain soybean"

Both have similar embeddings.

6. What is Chunking?

Answer:

Chunking means splitting a large document into smaller pieces for better retrieval.

7. Why is Chunking needed?

Answer:

Large documents cannot be processed efficiently at once, so they are divided into smaller chunks.

8. What is Semantic Search?

Answer:

Semantic search finds information based on meaning rather than exact keywords.

9. Which vector database did you use?

Answer:

I used FAISS.

10. Which embedding model did you use?

Answer:

I used Hugging Face embedding models such as BGE.

11. What is LangChain?

Answer:

LangChain is a framework used to build applications with LLMs, RAG systems, and chatbots.

12. What is a Retriever?

Answer:

A retriever fetches the most relevant document chunks from the vector database.

13. What is the workflow of your project?

Answer:

PDF

↓

Chunking

↓

Embeddings

↓

FAISS

↓
Retriever
↓
Answer

14. What is a Vector Database?

Answer:

A vector database stores embeddings and retrieves similar vectors quickly.

15. What is NLP?

Answer:

NLP (Natural Language Processing) enables computers to understand and process human language.

16. How does FAISS find relevant information?

Answer:

FAISS compares vector similarity between the question embedding and document embeddings.

17. What PDF did you use?

Answer:

I used a Soybean Production Guide PDF for testing the chatbot.

18. Can your chatbot answer in multiple languages?

Answer:

Yes, it supports multilingual queries such as English, Marathi, and Hindi.

19. What challenges did you face?

Answer:

- PDF file path issues
 - FAISS index loading errors
 - Embedding model downloads
 - Package installation problems
-

20. What did you learn from this project?

Answer:

I learned:

- NLP
- LangChain
- FAISS
- Embeddings
- Semantic Search
- Vector Databases
- RAG Architecture

21. What is the difference between Keyword Search and Semantic Search?

Answer:

Keyword Search	Semantic Search
Matches words	Matches meaning
Exact keywords required	Understands context
Less accurate	More accurate

22. Why did you use FAISS instead of SQL?

Answer:

FAISS is designed for vector similarity search, while SQL databases are designed for structured data.

23. What is the benefit of RAG?

Answer:

RAG provides document-based answers and reduces hallucinations.

24. How can you improve this project?

Answer:

- Add Ollama
- Add Llama 3
- Create a Streamlit UI
- Add chat history
- Support multiple PDFs

25. Explain your project in 30 seconds.

Answer:

I developed a Local RAG-based Question Answering System using Python, LangChain, Hugging Face Embeddings, and FAISS. The system processes PDF documents, converts them into embeddings, stores them in a vector database, and retrieves relevant information based on user queries. It supports semantic search and multilingual question answering.